

Conserved quadratic fluctuations and collisional drift in a hard-sphere gas

A conditional functional theorem and intrinsic-moment extension

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Abstract

For the three-dimensional excluded-product grand-canonical hard-sphere gas with diameter ε and activity $\mu = \varepsilon^{-2}$, we prove a functional Gaussian limit for the maximal thirteen-dimensional quadratic family closed under free transport and co-located elastic collision invariance. The limit contains an initial vacancy bias and a finite-diameter deterministic collisional drift; its Gaussian randomness is entirely the initial field. The proof supplies the two-root connected estimate for speed-weighted passive records, derives a uniform incidence law of large numbers, and uses exact mechanical identities rather than an unbounded-test functional theorem. A corollary treats per-particle thermal energy, intrinsic virial, and spread about the center of mass. The assertion is conditional on the explicitly stated, commit-pinned operational history package and the existing exactly centered Schwartz functional theorem. True-flow distributional convergence, first-moment transfer, and pasted prelimit covariance convergence are proved separately. The available probability coupling does not establish true prelimit covariance convergence, and does not imply its divergence. No full phase-space mean correction is asserted.

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1 Model and principal statements

Write $z = (x, v) \in \mathbb{R}^3 \times \mathbb{R}^3$ and $f_t(q) = \int q(z) f_t(z) dz$. The diameter is ε , the activity is $\mu = \varepsilon^{-2}$, and the initial law is

$$\mathbb{P}_\varepsilon(N, dZ_N) = \mathcal{Z}_\varepsilon^{-1} \frac{\mu^N}{N!} \prod_{j=1}^N f_0(z_j) \prod_{i < j} \mathbf{1}_{\{|x_i - x_j| > \varepsilon\}} dZ_N. \quad (1.1)$$

Assumption 1.1 (Datum and regular kinetic interval). The function f_0 is a smooth nonnegative probability density supported in $\{|x| \leq R_x, |v| \leq R_v\}$ for fixed finite R_x, R_v . On $[0, T]$ there is a nonnegative classical Boltzmann solution f_t with initial datum f_0 and a uniform bound $\sup_{t \leq T, x, v} e^{2\beta|v|^2} f_t(x, v) < \infty$ for some $\beta > 0$. The propagated Bol norms and global velocity moments of the pinned package in Section 2 are used.

For reference, the spatially summable Gaussian Bol norm is

$$\|g\|_{\text{Bol}, \beta} = \sum_{a \in \mathbb{Z}^3} \sup_{|x-a| \leq 1, v \in \mathbb{R}^3} e^{\beta|v|^2} |g(x, v)|.$$

Compact initial support is stronger than the datum assumption of entry 009. No compact velocity support of f_t at positive times is assumed.

For centers $x, x + \varepsilon\omega$ and incoming velocities v, w , use

$$B = ((v - w) \cdot \omega)_+, \quad v' = v - ((v - w) \cdot \omega)\omega, \quad w' = w + ((v - w) \cdot \omega)\omega. \quad (1.2)$$

Scattering occurs only in the incoming sector $B > 0$. The angular measure on S^2 is ordinary surface measure, and

$$\int Q(f, f) \phi dv = \frac{1}{2} \int B f f_* [\phi(v') + \phi(w') - \phi(v) - \phi(w)] dv dw d\omega.$$

In particular $\partial_t f + v \cdot \nabla_x f = Q(f, f)$.

For the true elastic flow and the single pasted flow described below, put

$$\pi_t^\varepsilon = \mu^{-1} \sum_{j=1}^N \delta_{z_j(t)}, \quad \zeta_t^\varepsilon(q) = \varepsilon^{-1} [\pi_t^\varepsilon(q) - \mathbb{E} \pi_t^\varepsilon(q)], \quad Y_t^\varepsilon(q) = \varepsilon^{-1} [\pi_t^\varepsilon(q) - f_t(q)]. \quad (1.3)$$

Superscript p denotes the pasted law. Velocities are outgoing and right continuous at contacts; positions are continuous. Thus the moment paths are càdlàg. The sphere normalization gives

$$J_f(s) = \int B^2 f_s(x, v) f_s(x, w) dx dv dw d\omega = \frac{2\pi}{3} \int |v - w|^2 f_s(x, v) f_s(x, w) dx dv dw, \quad (1.4)$$

$$H(t) = \frac{1}{2} \int_0^t J_f(s) ds.$$

The Gaussian pointwise bound on one factor and mass and energy on the other bound J_f integrably. Hence H is continuous and nondecreasing. Set

$$\rho_0(x) = \int f_0(x, v) dv, \quad b_{\text{ex}} = \frac{4\pi}{3}, \quad a(q) = -b_{\text{ex}} \int \rho_0(x) f_0(x, v) q(x, v) dx dv. \quad (1.5)$$

1.1 The maximal quadratic family

Let $\mathcal{Q}$ be the space of polynomials of total degree at most two for which $q(x + tv, v)$ is a co-located elastic collision invariant for every real t . It is the thirteen-dimensional space

$$q(x, v) = \alpha + b \cdot x + c \cdot v + \kappa |v|^2 + x^T R v + d x \cdot v + e |x|^2, \quad R^T = -R. \quad (1.6)$$

The coefficients $d = d(q)$ and $e = e(q)$ are uniquely determined. A basis is

$$1, \quad x_1, x_2, x_3, \quad v_1, v_2, v_3, \quad |v|^2, \quad (x \times v)_1, (x \times v)_2, (x \times v)_3, \quad x \cdot v, \quad |x|^2. \quad (1.7)$$

Write $r_q(t, z) = q(x + tv, v)$. Let Z_0 be a centered Gaussian linear map on $\mathcal{Q}$ with covariance

$$\mathbb{E} Z_0(q) Z_0(r) = \int f_0(z) q(z) r(z) dz. \quad (1.8)$$

There is no canonical subtraction in this grand-canonical covariance.

Theorem 1.2 (Conserved quadratic functional limit). *Assume Assumption 1.1, H1–H6 in Section 2, and the existing functional input F in that section. For either the true flow or the one fixed pasted flow, the vector of Y^ε on the basis (1.7) converges in $D([0, T], \mathbb{R}^{13})$, with its ordinary J_1 topology, to the continuous Gaussian process determined by*

$$Y_q(t) = Z_0(r_q(t)) + a(r_q(t)) + d(q)H(t) + 2e(q) \int_0^t H(s) ds, \quad q \in \mathcal{Q}. \quad (1.9)$$

The convergence is joint with the exactly centered strong tempered-distribution process of entry 009. If its Gaussian limit is G , then $Z_0(q) = G_0(\eta q)$ for any $\eta \in C_c^\infty$ equal to one near $\text{supp } f_0$.

For both laws the first moments satisfy, uniformly in t ,

$$\mathbb{E} Y_t^\varepsilon(q) \longrightarrow a(r_q(t)) + d(q)H(t) + 2e(q) \int_0^t H(s) ds. \quad (1.10)$$

Their exactly centered moment processes converge to $Z_0(r_q(t))$. The covariance of the limiting process is

$$\text{Cov}(Y_q(t), Y_r(s)) = \int f_0(z) r_q(t, z) r_r(s, z) dz. \quad (1.11)$$

For the pasted law its prelimit covariances converge uniformly in t, s to (1.11). True prelimit covariance convergence is not a conclusion of the present theorem.

The proof establishes that (1.6) is maximal with its stated transport/invariance property. A general anisotropic spatial quadratic does not share this scalar conserved closure.

Corollary 1.3 (Energy, virial, and spread). *For $q_E = |v|^2$, $q_V = x \cdot v$, and $q_X = |x|^2$, the limit in $D([0, T], \mathbb{R}^3)$ is*

$$Y(t) = A(t)(Z + a_*) + c(t), \quad A(t) = \begin{pmatrix} 1 & 0 & 0 \\ t & 1 & 0 \\ t^2 & 2t & 1 \end{pmatrix}, \quad c(t) = \begin{pmatrix} 0 \\ H(t) \\ 2 \int_0^t H(s) ds \end{pmatrix}, \quad (1.12)$$

where $(a_)_i = a(q_i)$ and $\text{Cov}(Z_i, Z_j) = \int f_0 q_i q_j$. Its covariance is $A(t) \Sigma A(s)^T$, and its mean is*

$$a(|v|^2), \quad a(x \cdot v + t|v|^2) + \frac{1}{2} \int_0^t J_f(s) ds, \quad a(|x + tv|^2) + \int_0^t (t-s) J_f(s) ds.$$

The scalar q_E equals twice the physical kinetic energy. The initial vacancy term and the collisional term are both part of Boltzmann centering.

2 Pinned operational hypotheses

The conditional boundary is explicit. Sources F and V mean the public entry-009 manuscript and virial supplement [1, 2] at

87868bcb65bf0460a90d6bc2efc3324e46481230.

Their upstream source U is [3] at

adc7f1241b42e322a6451854ab7e4b4c146bf78a.

The H1–H6 hypotheses below have the same mathematical content as those in V. They are imported, not independently reproved here. Appendix A maps them to exact labels and files.

2.1 One auxiliary dynamics

Choose the admissible cycle-excess cap Γ and fixed number L of slabs from U before later fixed moment orders. Put $\tau = T/L$ and, up to integer rounding,

$$A_L = |\log \varepsilon|, \quad \Lambda_i = A_i^{30}, \quad A_{i-1} = \Lambda_i^{30} \quad (i = L, \dots, 1). \quad (2.1)$$

An incoming contact is accepted when adding its edge keeps the connected collision component within Λ_i labels and excess Γ in slab i . Excess is edges minus vertices plus components; repeated contacts are repeated edges. Accepted C contacts scatter elastically; rejected contacts continue straight. Only collision prehistory restarts at slab interfaces. Positions and velocities are not reset, and overlaps are allowed. Expansion O contacts are signed non-scattering contacts and are distinct from accepted C events.

2.2 H1–H6

H1. Initial and kinetic control. Ordered factorial densities normalized by μ^{-h} obey $0 \leq F_{h,0}^\varepsilon \leq f_0^{\otimes h}$ and $(1 - C\varepsilon)f_0 \leq F_{1,0}^\varepsilon \leq f_0$. In particular $\sup_\varepsilon \mathbb{E}(N/\mu)^p < \infty$ for each fixed integer p . One-root birth inputs g_i have a common Gaussian Bol envelope and converge to $f_{i\tau}$ in a common smaller Bol norm, including a partial final slab. The kinetic bounds propagate and provide global velocity moments.

H2. Autonomous partition and output algebra. For a finite label set V , let Ψ_C be autonomous capped evolution on block C , κ_C its connected-accepted-graph indicator, and $\chi(C, D)$ the indicator that the two autonomous blocks have no cross-contact acceptable under their prefix prehistories. For designated roots S and a bounded root-history test K ,

$$1 = \sum_{C \in \mathcal{P}(V)} \prod_{C \in \mathcal{C}} \kappa_C \prod_{\{C, D\} \subset \mathcal{C}} \chi(C, D), \quad (2.2)$$

$$K(\Psi_V|_S) = \sum_{C \in \mathcal{P}(V)} \prod_C \kappa_C \prod_{\{C, D\}} \chi(C, D) K(\Psi_C|_S). \quad (2.3)$$

Expanding incompatibilities into rooted forest blocks gives a finite retained operator $\mathcal{L}_s$ and positive stopped domination $\mathcal{E}_s$ with $F'_s = \mathcal{L}_s F + R_s$ and $|R_s(K)| \leq (\mathcal{E}_s F)(|K|)$. The signed kernels are root-output pushforwards multiplied by physical indicators; their weights are

independent of passive records. Spectators sum to one before integration. Choosing j fresh labels contributes precisely $\mu^j/j!$, using

$$(n-s)!^{-1} \binom{n-s}{j} = [j!(n-s-j)!]^{-1}.$$

Required recorded lines continue to time zero. Forgetting records commutes with output pushforwards and one-root regrouping; newly unmarked singleton blocks sum to g_i . These are exact algebraic identities, not independence of the actual gas at interfaces.

H3. Unweighted remainder and complexity tails. For fixed factorial order and fixed finite bounded record count, the actual and retained factorial hierarchies satisfy

$$\|F_h^\varepsilon - \widehat{F}_h^\varepsilon\|_{\text{TV}} \leq C_h e^{-c_h \Lambda_L}, \quad \Lambda_L = |\log \varepsilon|^{30}, \quad (2.4)$$

also on partial final slabs. Finite factorial-cumulant inversion retains connected diagrams. Complexity is $\rho = |H_0| + \sum_i (|H_i| + r_i)$, with continuing/designated roots H_i , initial-link incident lines H_0 , and slab cycle excess r_i . At every fixed order and prescribed P , choosing a fixed K makes the retained sum with $\rho > K$ be $O(\varepsilon^P)$. Nonphysical initial tails also admit an arbitrary-power auxiliary choice. These choices do not alter the dynamics. No energy or Gaussian weight is assumed for a general microscopic remainder.

H4. Chronological coordinates and positive retained bounds. Every global birth has one independent phase coordinate; continuation adds none. If E is total birth energy, active energy never exceeds E . Mergers of disjoint chronological components have (s, ω) coordinates with Jacobian $\varepsilon^2 B \, ds \, d\omega$. A redundant contact is a determined incoming-intersection indicator with update fixed by preceding data; a missing branch is zero. At fixed complexity, let m be birth count, q continuation count, p physical initial-link count, $n = \sum_i n_i = m + q$, and $a = \sum_i a_i = m - 1 - p$ for a connected one-root history. Then

$$n - a = q + 1 + p = O_K(1), \quad \mu^{m-1} \varepsilon^{2a} \varepsilon^{3p} = \varepsilon^p.$$

There are $O_K(1)$ redundant contacts. Integrating one common translation, or prescribing a root position and taking spatial suprema, bounds the birth product by $C^m e^{-bE}$. Fresh-label factorials, ordered times, and pair choices give the size majorant

$$C_K (1+n)^{D_K} (C\tau\sqrt{L})^n, \quad C\tau\sqrt{L} < 1. \quad (2.5)$$

It holds globally and at a fixed root position with a fixed small Gaussian reservation. The exponential base and choice of L are independent of later fixed polynomial degrees and moment orders. For h designated roots the connected normalization is $\mu^{1-h} \varepsilon^p$, as follows from the same birth count; see (3.6) below and F's imported package.

H5. Finite-tree convergence, completeness, and splitting. At fixed finite size and bounded birth energy, prescribed paths, times, normals, and velocities converge as $\varepsilon \rightarrow 0$ away from the null grazing, zero-relative-speed, coincident-time, and extra-contact sets. There are no undeclared contacts for sufficiently small ε . Physical initial links retain their ε factor. Cyclic histories vanish qualitatively after truncation and domination. The surviving signed C/O trees form the complete leading class, with signs $(-1)^{\#O}$ and inputs $f_{i\tau}$; their unmarked one-root marginal at time t is f_t .

After unused records are projected out, reverse-chronological C/O cancellation removes the latest contacts meeting no tested root. Deleting the last remaining root contact splits its tree into two incoming components at their common position, with fresh-label factorization

$$(j_1 + j_2)!^{-1} \binom{j_1 + j_2}{j_1} = (j_1! j_2!)^{-1}.$$

This operation is made in the finite limiting tree class with a combined-size cutoff, removed subsequently by absolute summability. Conditional spatial tree measures are weakly continuous in root position with uniform conditional and integrated Gaussian bounds. Their equal-position collision product is defined by the common-translation coordinates, equivalently by an approximate identity.

H6. Full-time true/pasted coupling. On common initial data, the probability of a trajectory discrepancy anywhere on $[0, T]$ is at most $C\varepsilon^{21/20}$. The almost-everywhere elastic flows and their outgoing right-continuous velocity versions are defined, and deterministic-time collision sets are null.

2.3 Existing functional input F

We use the already proved exactly centered functional theorem of F: the true and the one fixed pasted signed-measure fields converge weakly in $D([0, T], \mathcal{S}'_{\beta}(\mathbb{R}^6))$, with the generalized J_1 topology induced by the strong Schwartz dual, to a Radon law supported on strongly continuous paths. Its finite Schwartz evaluations are centered Gaussian with covariance

$$\mathbb{E}G_t(\phi)G_s(\psi) = \int f_0 U(0, t)\phi U(0, s)\psi + \int_0^{s \wedge t} C_r(U(r, t)\phi, U(r, s)\psi) dr. \quad (2.6)$$

Here U is the existing backward propagator, and C_r is its collision noise covariance. Initial covariance is $\int f_0 \phi \psi$ and the noise is independent of the initial field. The static initial bounded-test CLT and convergence of its covariances are used in their existing scope. For clarity, the pinned initial proof gives $\mu \|c_{2,0}^\varepsilon\|_{TV} = O(\varepsilon)$. No unbounded positive-time test CLT and no polynomial mapping property of U is included in input F. The source's nuclear tightness proof also supplies tightness of both path-law families along every sequence $\varepsilon_n \rightarrow 0$. This is the tightness used in the product argument in Section 5.

3 Speed records and a uniform incidence law

Give each particle a passive mark $y_j(t)$, initially zero. At every accepted C event add its incoming normal speed B to each participant. O and rejected contacts add zero. A particle never copies its partner's previous mark. Define the random cumulative speed sum

$$\mathcal{H}_\varepsilon^p(t) = \frac{1}{2\mu} \sum_j y_j(t) = \frac{1}{\mu} \sum_{c \in C_t^p} B_c. \quad (3.1)$$

Its expectation is denoted $m_\varepsilon(t)$. This random variable must be distinguished from V's deterministic notation for its expectation.

With root phase/record variables ξ_j , let

$$F_h^\varepsilon = \mu^{-h} \mathbb{E} \sum_{i_1, \dots, i_h \text{ distinct}} \delta_{(\xi_{i_1}, \dots, \xi_{i_h})}, \quad c_1^\varepsilon = F_1^\varepsilon, \quad c_2^\varepsilon = F_2^\varepsilon - F_1^\varepsilon \otimes F_1^\varepsilon.$$

Records at different times can be included as a fixed finite root-output list. No positive separation of those times is needed.

Lemma 3.1 (Weighted one- and two-root bounds). *Uniformly over observation times in $[0, T]$, for $h \leq 2$ and nonnegative integers b_j with $\sum b_j \leq 2$,*

$$\left| \int c_h^{\varepsilon, p} \prod_{j=1}^h y_j^{b_j} \right| \leq C\mu^{1-h}. \quad (3.2)$$

In particular

$$\sup_{t \leq T} \text{Var}(\mathcal{H}_\varepsilon^p(t)) \leq C\mu^{-1} = C\varepsilon^2. \quad (3.3)$$

Proof. We first bound actual records, then estimate all signed-branch tests globally. In slab i a final accepted component has at most Λ_i labels and no accepted edge to its complement. Its energy is conserved throughout the slab, including before its prefixes merge. If slab-start speeds are bounded by V_i , every later speed in that component is at most $\sqrt{\Lambda_i}V_i$. Across the fixed slab list this gives

$$V_\varepsilon = R_v \prod_{i=1}^L \sqrt{\Lambda_i}, \quad \max_{j,t} |v_j^p(t)| \leq V_\varepsilon.$$

Its accepted multigraph has at most $\Lambda_i - 1 + \Gamma$ edges, so

$$0 \leq y_j(t) \leq D_\varepsilon, \quad D_\varepsilon = 2V_\varepsilon \sum_i (\Lambda_i - 1 + \Gamma), \quad P_\varepsilon = 1 + D_\varepsilon. \quad (3.4)$$

For fixed L these quantities are fixed powers of $1 + |\log \varepsilon|$.

Append the passive coordinate to the H2 root-output maps and use $\ell_\varepsilon(y) = \text{clip}(y/P_\varepsilon, -1, 1)$ on *every* signed branch. Compatibility reproduces the same accepted history, hence the same mark. Kernel weights, spectators, fresh-label coefficients, and one-root regrouping are unchanged. Pushforward has variation norm at most one. Required recorded lines continue to zero. Thus the finite output algebra and H3 unweighted error mechanism remain valid, just as for the accepted-contact count record in F. The mark normalization changes no trajectory.

On the actual process $P_\varepsilon \ell_\varepsilon(y) = y$, while on any artificial branch $|P_\varepsilon \ell_\varepsilon(y)| \leq |y|$. The bounded test must be global: actual speed support cannot be assigned to artificial independent birth inputs. Write $\delta_h = \|F_h - \hat{F}_h\|_{\text{TV}}$. Actual factorial masses are at most one by H1 and conservation of N , so $\|\hat{F}_1\|_{\text{TV}} \leq 1 + \delta_1$. Finite inversion gives, directly,

$$\|c_2 - \hat{c}_2\|_{\text{TV}} \leq \delta_2 + \delta_1(2 + \delta_1).$$

Restoring degree at most two costs at most P_ε^2 . It leaves H3's exponential remainder smaller than every fixed power of ε . Choose complexity and nonphysical initial-tail accuracies after this degree; for example an unweighted $O(\varepsilon^6)$ tail, restored by a fixed polylogarithm, is $o(\varepsilon^2) = o(\mu^{-1})$. For a later field covariance, its additional factor μ is included before choosing accuracy. No weighted microscopic remainder is used.

At fixed complexity K , total slab size n , and total birth energy E , there are at most $n + C_K$ prescribed contacts and $B \leq \sqrt{2E}$. Consequently

$$|y_j| \leq C_K(1+n)\sqrt{E}, \quad \left| \prod_j y_j^{b_j} \right| \leq C_K(1+n)^2(1+E) \quad \text{if } \sum b_j \leq 2. \quad (3.5)$$

This uses active energy on the realizable branch, including determined redundant updates; unrealizable branches remain zero. It also dominates the restored clipped output. With m births, $a = m - 1 - p$ mergers, and p physical initial links, the exact connected normalization at h roots is

$$\mu^{m-h} \varepsilon^{2a} \varepsilon^{3p} = \mu^{1-h} \varepsilon^p. \quad (3.6)$$

Continuation contributes no extra birth or activity factor.

We detail why the energy insertion preserves the size base. On a dyadic energy bin of upper scale P , summing the next merger pair in slab i costs at most $n_i^{3/2} \sqrt{P}$. Normal integration adds a fixed exponential factor, ordered merger times give $\tau^{a_i}/a_i!$, and fresh labels give $1/n_i!$ up to fixed complexity polynomials. The birth product after one translation integration is $C^m e^{-bE}$. The source's energy-bin calculation therefore gives, before the final Stirling estimate,

$$C^{m+a} \tau^a (1+a)^{a/2} \prod_i \frac{n_i^{3a_i/2}}{a_i! n_i!}, \quad (3.7)$$

up to a fixed polynomial. Reserving a fixed Gaussian fraction absorbs $(1+E)$ in (3.5). Equivalently, a fixed integer shift r of the energy-bin power has gamma ratio

$$\frac{\Gamma(s+r+1)}{\Gamma(s+1)} = \prod_{k=1}^r (s+k) \leq C_r (1+s)^r,$$

and costs only a fixed power of size. Since $\delta_i = n_i - a_i \geq 0$ and $\sum_i \delta_i = O_K(1)$, Stirling and convexity yield

$$\frac{n_i^{3a_i/2}}{a_i! n_i!} \leq e^{2n_i} (1+n_i)^{\delta_i} n_i^{-n_i/2}, \quad \prod_i n_i^{n_i} \geq (n/L)^n.$$

Inserting these into (3.7) gives $C_K \mu^{1-h} \varepsilon^p (1+n)^{D_K \theta^n}$ with $\theta = C\tau\sqrt{L} < 1$. Choose L once for the fixed smaller Gaussian exponent. Later fixed degrees change front constants and powers of size, not that choice. Summing n and adding the restored small errors proves (3.2). The reasoning is uniform on partial slabs and finite record lists.

Finally, for $S_y = \sum_j y_j$, the identical-index and distinct-index terms give the exact identity

$$\text{Var } S_y = \mu \int F_1 y^2 + \mu^2 \int c_2 y_1 y_2.$$

Dividing by $4\mu^2$ proves (3.3). The second integral may be signed; its absolute bound suffices. The $1/2$ in (3.1) is necessary because each contact is recorded by both participants. $\square$

Lemma 3.2 (Speed-incidence mean). *The same pasted dynamics satisfies $m_\varepsilon(t) = \mathbb{E}\mathcal{H}_\varepsilon^p(t) \rightarrow H(t)$ uniformly on $[0, T]$.*

Proof. This is the weighted incidence theorem of V; the short proof from the stated package is included. Remove microscopic and complexity errors with the globally clipped degree-one test as above. At fixed size and bounded birth energy the raw finite speed sum converges by H5, and clipping is eventually inactive. The bound (3.5), its Gaussian domination, and the size majorant remove the energy and size cutoffs. Physical initial links retain ε and cyclic terms vanish after the same weighted truncation and domination. Bol-norm convergence replaces interface births by $f_{i\tau}$. No qualitative normalized-record convergence is multiplied by P_ε .

In the surviving finite limiting tree class, inspect contacts in reverse chronological order. Project unused future auxiliary records. The latest contact that meets no tested root has equal incoming

C/O flux, opposite signs, and the same root output, including its mark, so its choices cancel. Repeat until the latest remaining contact meets the root. Delete that contact at time s using H5. Its two incoming components factor with their original fresh-label weights and the combined-size cutoff. For the terminal test y , the C-minus-O increment is exactly B , independently of both incoming historical marks. Both components can therefore be projected onto their unmarked phase points, whose measures are $f_s(x, v)$ and $f_s(x, w)$. The one-root moment is

$$M_y(t) = \int_0^t \int B^2 f_s(x, v) f_s(x, w) dx dv dw ds.$$

The flux supplies one B and the mark supplies the second. Absolute summability removes the combined-size cutoff before replacing the incoming components by their unrestricted product. A one-root insertion has no factor $1/2$; the relation $\sum_j y_j = 2 \sum_c B_c$ then gives $H = M_y/2$. The unused partner history was projected out, not replaced by a fresh zero mark. The argument holds at every fixed time, including a partial final slab. Both m_ε and H are nondecreasing and H is continuous. A fixed finite grid with small H increments sandwiches intermediate values, giving uniform convergence from the fixed-time convergence. $\square$

Proposition 3.3 (Uniform incidence LLN). *For the pasted flow,*

$$\mathbb{E} \sup_{t \leq T} |\mathcal{H}_\varepsilon^p(t) - H(t)|^2 \longrightarrow 0. \quad (3.8)$$

For the true speed sum $\mathcal{H}_\varepsilon(t) = \mu^{-1} \sum_{c \in \mathcal{C}_t} B_c$,

$$\sup_{t \leq T} |\mathcal{H}_\varepsilon(t) - H(t)| \longrightarrow 0 \quad \text{in probability.} \quad (3.9)$$

Proof. Given $\eta > 0$, take a fixed grid whose successive H increments are at most η . Uniform mean convergence makes successive m_ε grid increments at most 3η for small ε . Every sample and every mean $\mathcal{H}_\varepsilon^p, m_\varepsilon$ is nondecreasing. Thus

$$\sup_t |\mathcal{H}_\varepsilon^p(t) - m_\varepsilon(t)| \leq \max_i |\mathcal{H}_\varepsilon^p(t_i) - m_\varepsilon(t_i)| + 3\eta.$$

The squared expectation of the maximum is bounded by the sum of its endpoint variances, at most $C \#\{t_i\} \varepsilon^2$. First let $\varepsilon \rightarrow 0$ for this fixed grid, then $\eta \rightarrow 0$, and add $\|m_\varepsilon - H\|_\infty$. This proves (3.8). On the good event in H6 the true and pasted events, hence their entire speed-sum paths, agree. Its exceptional probability tends to zero, proving (3.9). No squared bad-event comparison is used. $\square$

4 Conservation, maximality, and initial bias

Lemma 4.1 (Mechanical identities and global quadratic bounds). *For either microscopic flow, almost surely on its defined trajectories,*

$$\pi_t^\varepsilon(|v|^2) = \pi_0^\varepsilon(|v|^2), \quad (4.1)$$

$$\pi_t^\varepsilon(x \cdot v) = \pi_0^\varepsilon(x \cdot v) + t \pi_0^\varepsilon(|v|^2) + \varepsilon \mathcal{H}_\varepsilon(t), \quad (4.2)$$

$$\pi_t^\varepsilon(|x|^2) = \pi_0^\varepsilon(|x|^2) + 2t \pi_0^\varepsilon(x \cdot v) + t^2 \pi_0^\varepsilon(|v|^2) + 2\varepsilon \int_0^t \mathcal{H}_\varepsilon(s) ds. \quad (4.3)$$

Mass, momentum, and angular momentum are constant, and the position sum is its initial value plus t times momentum. For every fixed $q \in \mathcal{Q}$,

$$\sup_{t \leq T} \sum_j |q(z_j(t))| \leq C_{q,T} N. \quad (4.4)$$

Proof. Energy conservation and the initial velocity support give $\|V(t)\|_2 \leq R_v \sqrt{N}$. With continuous positions and their piecewise velocity derivative, Minkowski gives

$$\|X(t)\|_2 \leq \|X(0)\|_2 + \int_0^t \|V(s)\|_2 ds \leq (R_x + TR_v) \sqrt{N}.$$

Cauchy–Schwarz controls the absolute mixed and angular sums and the linear position and velocity sums, proving (4.4). At a contact the pair energy and momentum jumps are zero, whereas

$$x \cdot (v' - v) + (x + \varepsilon \omega) \cdot (w' - w) = \varepsilon B.$$

The angular jump is $\varepsilon B \omega \times \omega = 0$. Streaming gives derivatives $|v|^2$ for $x \cdot v$, $2x \cdot v$ for $|x|^2$, and zero for $x \times v$. Integration proves the identities. Accepted pasted contacts have exactly the same separation and orientation; rejected contacts and restarts have no phase impulse. Overlaps introduce no additional term. The nonnegative cumulative speed sum is bounded by $C_T N / \varepsilon$, so no extra true collision-count moment estimate is required for its integrability. H1 justifies taking expectations. $\square$

Proposition 4.2 (Maximal transport-closed quadratic invariants). *The definition of $\mathcal{Q}$ gives exactly (1.6). In spatial dimension d , its dimension is $2d + 4 + d(d - 1)/2$.*

Proof. Write a general quadratic as

$$\alpha + b \cdot x + c \cdot v + x^T D x + x^T M v + v^T F v, \quad D^T = D, \quad F^T = F.$$

At a common collision position the linear-in-velocity terms depend only on pair momentum. Expressing the velocities by their center velocity and relative velocity g , pair invariance of the remaining term requires $g^T F g$ invariant under every reflection of g . A reflection can send a vector to any other vector of the same length; hence $F = \kappa I$. The velocity-quadratic coefficient of $q(x + tv, v)$ is $F + t \operatorname{sym} M + t^2 D$. For it to be scalar at every t , $\operatorname{sym} M = dI$ and $D = eI$. The skew part of M is arbitrary. Conversely these conditions make all the transported tests collision invariant. They give (1.6). Counting its independent coefficients gives the dimension. This proves maximality with the stated property, without asserting invariance for an arbitrary anisotropic quadratic. $\square$

For $q \in \mathcal{Q}$, the mechanical identities combine to

$$\pi_t^\varepsilon(q) = \pi_0^\varepsilon(r_q(t)) + \varepsilon \left[d(q) \mathcal{H}_\varepsilon(t) + 2e(q) \int_0^t \mathcal{H}_\varepsilon(s) ds \right]. \quad (4.5)$$

The skew term in (1.6) has neither a streaming derivative nor a finite-diameter impulse. Boltzmann's co-located invariants give

$$f_t(q) = f_0(r_q(t)). \quad (4.6)$$

These deterministic moment identities can be justified first with spatial and velocity cutoffs. Global energy and the propagated spatial second moment remove them; the latter follows from compact initial support and $2|x \cdot v| \leq |x|^2 + |v|^2$. This deterministic integration is not an unbounded fluctuation theorem.

Lemma 4.3 (Initial vacancy bias). *For every fixed polynomial q ,*

$$a_\varepsilon(q) := \varepsilon^{-1} [\mathbb{E} \pi_0^\varepsilon(q) - f_0(q)] \longrightarrow a(q). \quad (4.7)$$

Proof. The insertion identity for (1.1) is $F_{1,0}^\varepsilon(x, v) = f_0(x, v)\mathbb{P}\{K_x = 0\}$, where K_x counts centers of an independent initial configuration in the radius- ε ball about x . H1, smoothness of ρ_0 , and $\mu\varepsilon^3 = \varepsilon$ give, uniformly in x ,

$$\mathbb{E}K_x = \varepsilon b_{\text{ex}}\rho_0(x) + O(\varepsilon^2), \quad \mathbb{E}K_x(K_x - 1) \leq \mu^2 \left(\int_{B(x, \varepsilon)} \rho_0 \right)^2 = O(\varepsilon^2).$$

The elementary Bonferroni inequalities

$$\mathbb{E}K_x - \frac{1}{2}\mathbb{E}K_x(K_x - 1) \leq \mathbb{P}(K_x \geq 1) \leq \mathbb{E}K_x$$

yield $F_{1,0}^\varepsilon = f_0[1 - \varepsilon b_{\text{ex}}\rho_0 + O(\varepsilon^2)]$. Multiply by q and integrate on the compact support of f_0 . This proves (4.7). In particular the activity μ cannot be identified with $\mathbb{E}N$ at the order relevant to Boltzmann centering. $\square$

Lemma 4.4 (True mean transfer). *For $q \in \mathcal{Q}$ and every fixed integer $p > 21$,*

$$\sup_{t \leq T} \varepsilon^{-1} |\mathbb{E}\pi_t^\varepsilon(q) - \mathbb{E}\pi_t^{\varepsilon,p}(q)| \leq C_{q,T,p} \varepsilon^{(p-21)/(20p)}. \quad (4.8)$$

Moreover $\mathbb{E}\mathcal{H}_\varepsilon \rightarrow H$ uniformly.

Proof. Let E_ε be H6's full-time discrepancy event. Off it the moment paths agree; on it (4.4) bounds their absolute particle-sum difference by $C_{q,T}N$. Holder and H1 give

$$\varepsilon^{-1} \mathbb{E}[(N/\mu)\mathbf{1}_{E_\varepsilon}] \leq C_p \varepsilon^{-1+(21/20)(1-1/p)} = C_p \varepsilon^{(p-21)/(20p)}.$$

For $p = 42$ the exponent is $1/40$. Apply (4.2) to both coupled flows: their initial and energy terms agree, so $\mathcal{H}_\varepsilon - \mathcal{H}_\varepsilon^p = \varepsilon^{-1}[\pi_t^\varepsilon(x \cdot v) - \pi_t^{\varepsilon,p}(x \cdot v)]$. The mean comparison therefore controls the speed-sum expectation uniformly. Lemma 3.2 completes the proof. This is first-moment transfer, not a deduction from probability coupling alone. $\square$

5 Proof of the functional and joint limits

Proof of Theorem 1.2. Fix $\eta \in C_c^\infty$ equal to one near $\text{supp } f_0$. Initially every particle lies in that support, so for each $q \in \mathcal{Q}$, $\pi_0^\varepsilon(q) = \pi_0^\varepsilon(\eta q)$ almost surely, including equality of expectations. Input F therefore gives the initial Gaussian vector Z_0 with covariance (1.8). This is an exact compact-test substitution at time zero, not an unbounded positive-time observation. Lemma 4.3 adds its deterministic vector a for Boltzmann centering. Since $r_q(t)$ is a fixed finite linear combination of the basis (1.7), all of its initial fluctuations follow jointly from this one vector.

Subtract (4.6) from (4.5) and divide by ε :

$$Y_t^\varepsilon(q) = Y_0^\varepsilon(r_q(t)) + d(q)\mathcal{H}_\varepsilon(t) + 2e(q) \int_0^t \mathcal{H}_\varepsilon(s) ds. \quad (5.1)$$

For the pasted law use (3.8); for the true law use (3.9). The event term converges uniformly in probability to its continuous deterministic limit. The initial-vector map is continuous into the space of continuous finite-coefficient polynomial paths. Ordinary finite-dimensional Slutsky gives the joint $D([0, T], \mathbb{R}^{13})$, J_1 limit (1.9). Indeed (5.1) provides a uniform-in-time residual tending to zero. No independence between the initial vector and the speed sum is required, since the latter limit is deterministic.

Taking expectations in the same identity, using the vacancy bias and the separate mean convergence of both speed sums, proves (1.10). Uniformity follows from the fixed polynomial coefficients and the uniform expectation limit. Subtracting the convergent means proves the exactly centered limit. Formula (1.11) is then the covariance of the Gaussian linear initial field.

For the pasted prelimit covariance, let $Z_{0,\varepsilon}(q) = \zeta_0^\varepsilon(q)$ and $R_{q,\varepsilon}(t) = d(q)[\mathcal{H}_\varepsilon^p(t) - m_\varepsilon(t)] + 2e(q) \int_0^t [\mathcal{H}_\varepsilon^p(s) - m_\varepsilon(s)] ds$. The centered path identity reads

$$\zeta_t^{\varepsilon,p}(q) = Z_{0,\varepsilon}(r_q(t)) + R_{q,\varepsilon}(t).$$

Proposition 3.3 implies $\mathbb{E} \sup_t |R_{q,\varepsilon}(t)|^2 \rightarrow 0$. The initial covariance matrices converge to (1.8) by the initial static factorial bounds, and are bounded. Finite coefficients in $r_q(t)$ and Cauchy–Schwarz make every covariance involving a residual uniformly small in t, s . The remaining initial covariance converges uniformly to (1.11). This argument transfers no squared bad-event error to the true law.

It remains to justify the product with the strong-dual field. Let $\mathfrak{E} = D([0, T], \mathcal{S}'_\beta)$ have input F's generalized J_1 topology, and let $\mathcal{Z}_\varepsilon$ be its exactly centered field. Evaluation at zero against a fixed finite Schwartz list is continuous: every allowed time change fixes zero. With the basis $q_1, \dots, q_{13}$,

$$Y_0^\varepsilon(q_i) = \mathcal{Z}_\varepsilon(0)(\eta q_i) + a_\varepsilon(q_i).$$

Define the continuous map from $\mathfrak{E}$ to $\mathfrak{E} \times D([0, T], \mathbb{R}^{13})$ which retains its first coordinate and constructs its second by (1.9), replacing $Z_0(q_i)$ by $z(0)(\eta q_i)$. Input F and continuous mapping give its weak limit. The actual second coordinate differs uniformly in probability from this construction by (5.1) and $a_\varepsilon \rightarrow a$.

For completeness, this perturbation step can be proved without assuming a general nonmetrizable Slutsky theorem. Given a bounded continuous function g on the product and a compact $K \subset \mathfrak{E}$ carrying all but arbitrarily small probability of the tight field laws, the constructed graph over K is compact. At each graph point continuity gives a field neighborhood and a radius in the metric J_1 second factor where g changes by less than a chosen δ . Shrink the field neighborhood using continuity of the constructed second coordinate. A finite subcover of K gives a common positive radius. If the field lies in K and the second-coordinate perturbation is smaller than this radius, the two g values differ by at most 2δ . Its exceptional probability tends to zero. Outside K , use $2\|g\|_\infty$; then let the compact exception and δ tend to zero. This proves convergence of all bounded continuous product tests, as required. The joint conclusion follows, with $Z_0(q) = G_0(\eta q)$. $\square$

Proof of Corollary 1.3. The transported tests are $r_E = |v|^2$, $r_V = x \cdot v + t|v|^2$, and $r_X = |x + tv|^2$. Their initial-coefficient matrix is $A(t)$ in (1.12). For virial $d = 1, e = 0$, and for spread $d = 0, e = 1$; energy has both zero. Theorem 1.2 gives all assertions. Fubini gives $2 \int_0^t H(s) ds = \int_0^t (t - s) J_f(s) ds$. $\square$

The joint Gaussian cross covariance, for $q \in \mathcal{Q}$ and $\phi \in \mathcal{S}$, is

$$\text{Cov}(G_s(\phi), Y_q(t)) = \int f_0(z) [U(0, s)\phi](z) r_q(t, z) dz. \quad (5.2)$$

Replace r_q by ηr_q on the compact initial support and use independence of the initial Gaussian field and its noise in input F. This proves (5.2) using only the existing compact/Schwartz interpretation of U . The phase-space field in the joint theorem retains exact microscopic centering; no generic Boltzmann recentering of that field is implicit.

The collisional drift is nonzero for the stated smooth nonzero datum. At some open set of positions the initial velocity density is positive on an open velocity set, so $\int |v - w|^2 f_0(x, v) f_0(x, w) dx dv dw > 0$. Continuity on a compact subregion preserves a positive lower bound for small positive times. Thus $H(t) > 0$ there. The Gaussian part, by contrast, comes entirely from the initial field because all transported tests in $\mathcal{Q}$ are co-located collision invariants.

6 Intrinsic per-particle moments

The conserved family has a nonlinear consequence without enlarging the unbounded-test class. Let

$$M_\varepsilon = \pi_0^\varepsilon(1) = N/\mu, \quad p_\varepsilon = \pi_t^\varepsilon(v)/M_\varepsilon, \quad x_\varepsilon(t) = \pi_t^\varepsilon(x)/M_\varepsilon.$$

On $N > 0$ define the intensive intrinsic moments

$$\begin{aligned} U_\varepsilon &= \pi_t^\varepsilon(|v|^2)/M_\varepsilon - |p_\varepsilon|^2, \\ D_\varepsilon(t) &= \pi_t^\varepsilon(x \cdot v)/M_\varepsilon - x_\varepsilon(t) \cdot p_\varepsilon, \\ S_\varepsilon(t) &= \pi_t^\varepsilon(|x|^2)/M_\varepsilon - |x_\varepsilon(t)|^2. \end{aligned} \tag{6.1}$$

Assign any fixed values on $N = 0$. The mass coordinate of Theorem 1.2 implies $M_\varepsilon \rightarrow 1$ in probability, so that exceptional probability tends to zero. Write

$$p_0 = f_0(v), \quad x_0 = f_0(x), \quad u_0 = f_0(|v - p_0|^2), \quad d_0 = f_0((x - x_0) \cdot (v - p_0)), \quad s_0 = f_0(|x - x_0|^2).$$

Their Boltzmann values are $(u_0, d_0 + tu_0, s_0 + 2td_0 + t^2u_0)$. Define three effective initial tests

$$\phi_U = |v - p_0|^2 - u_0, \quad \phi_D = (x - x_0) \cdot (v - p_0) - d_0, \quad \phi_S = |x - x_0|^2 - s_0. \tag{6.2}$$

Each belongs to $\mathcal{Q}$ and integrates to zero against f_0 .

Corollary 6.1 (Thermal energy and intrinsic spread). *For either flow, the Boltzmann-centered vector of (6.1), scaled by ε^{-1} , converges in $D([0, T], \mathbb{R}^3)$, J_1 , to*

$$A(t)(Z^{\text{int}} + a^{\text{int}}) + c(t), \quad \Sigma_{ij}^{\text{int}} = \int f_0 \phi_i \phi_j, \quad a_i^{\text{int}} = a(\phi_i), \tag{6.3}$$

where Z^{int} is centered Gaussian with covariance Σ^{int} , and A, c are from (1.12). This convergence is joint with the exactly centered Schwartz field, with $Z_i^{\text{int}} = G_0(\eta \phi_i)$. It describes the weak limit and its covariance; it does not add a true prelimit covariance assertion.

Proof. Mass and momentum are constant, and $x_\varepsilon(t) = x_\varepsilon(0) + tp_\varepsilon$. Substitution in (4.1)–(4.3) gives, exactly,

$$\begin{aligned} U_\varepsilon(t) &= U_\varepsilon(0), \\ D_\varepsilon(t) &= D_\varepsilon(0) + tU_\varepsilon(0) + \varepsilon \mathcal{H}_\varepsilon(t)/M_\varepsilon, \\ S_\varepsilon(t) &= S_\varepsilon(0) + 2tD_\varepsilon(0) + t^2U_\varepsilon(0) + 2\varepsilon \int_0^t \mathcal{H}_\varepsilon(s) ds/M_\varepsilon. \end{aligned}$$

Since $M_\varepsilon \rightarrow 1$, Proposition 3.3 gives the same uniform deterministic event term after division by M_ε .

For clarity, denote the raw initial moments by (M, P, X, E, V, Q) . The three rational maps are $E/M - |P|^2/M^2$, $V/M - X \cdot P/M^2$, and $Q/M - |X|^2/M^2$. At the deterministic point $(1, p_0, x_0, f_0(|v|^2), f_0(x \cdot v), f_0(|x|^2))$ their differentials, represented as tests of an empirical perturbation, are exactly (6.2). For example the first differential is

$$\delta E - 2p_0 \cdot \delta P + (2|p_0|^2 - f_0(|v|^2))\delta M = \delta \pi(|v - p_0|^2 - u_0).$$

The same calculation gives the other two tests. The initial thirteen-vector has an $O_{\mathbb{P}}(\varepsilon)$ deviation and its scaled limit is its Gaussian vector plus the vacancy bias. On M bounded away from zero the rational maps have bounded second derivatives in a fixed neighborhood. Thus their ε^{-1} -scaled Taylor remainders are $O_{\mathbb{P}}(\varepsilon) \rightarrow 0$. The finite-dimensional delta method gives the initial limit $Z^{\text{int}} + a^{\text{int}}$, jointly with the field. The exact triangular identities and the uniform event LLN prove (6.3). $\square$

The observables are unchanged by a constant spatial translation or Galilean boost. Normalization by the particle count removes the leading mass direction; this explains the zero means of the effective tests under f_0 , while the initial vacancy functional $a(\phi_i)$ can still be nonzero. The corollary does not require a separate positive-time CLT for these rational observables.

7 Endpoint cutoffs and precise scope

The main theorem needs only initial compact tests and exact conservation. The following separate endpoint gate records what Gaussian birth energy also proves, and the order of limits it permits.

Lemma 7.1 (Pasted polynomial endpoint covariance gate). *Let q be any fixed polynomial, $0 \leq \chi \leq 1$ a smooth cutoff equal to one on the unit phase ball and zero outside the radius-two ball, $\chi_R(z) = \chi(z/R)$, and $q^{>R} = q(1 - \chi_R)$. For every fixed $M > 0$, uniformly for sufficiently large R and sufficiently small ε ,*

$$\sup_{t \leq T} \mathbb{E} |\zeta_t^{\varepsilon, p}(q^{>R})|^2 \leq C_q e^{-c_q R^2} + C_{q, M} \varepsilon^M. \quad (7.1)$$

The statement holds for fixed finite endpoint record lists and gives covariance cutoff approximation at two times. In particular its cutoff limit takes $\varepsilon \rightarrow 0$ first, then $R \rightarrow \infty$.

Proof. The actual pasted speed bound (3.4) gives $|x_j^p(t)| \leq R_x + TV_\varepsilon$. Choose a fixed polylogarithmic $Q_\varepsilon \geq 1$ dominating $|q|$ on that actual phase region, independently of R . Use the globally bounded test $\text{clip}(q^{>R}/Q_\varepsilon, -1, 1)$ on every signed branch. On actual endpoints restoring Q_ε gives the raw tail, and on artificial branches its absolute value is bounded by $|q^{>R}|$. Restore degree two only after H3's unweighted remainder and tail estimates. Choose their arbitrary-power accuracy after the degree, requested M , and the covariance factor μ . All restored errors are $O(\varepsilon^M)$ uniformly in R .

Every tested endpoint is on a required record-bearing line continued to zero. Its initial input is supported in the compact position support of f_0 , either through $F_{1,0}^\varepsilon$ or through an initial cumulant dominated by products of f_0 . Retain this anchor before spatial envelope domination. H4 gives, on every nonzero retained branch with birth energy E ,

$$|v_j(t)| \leq \sqrt{E}, \quad |x_j(t)| \leq R_x + T\sqrt{E}.$$

The continued line is position-continuous. Thus for $R \geq \max(1, 2R_x)$ a nonzero tail factor forces $E \geq R^2/[4(T+1)^2]$. If q has degree d , two raw endpoint factors are bounded by

$$C_{q, T}(1 + E)^d \mathbf{1}_{\{E \geq c_T R^2\}}.$$

Only after extracting this implication discard the compact-anchor indicator. One birth factor is integrated in the common translation and the others bounded by spatial suprema, leaving $C^m e^{-bE}$ as in H4. One fixed Gaussian fraction extracts $e^{-c_q R^2}$, a second absorbs the fixed polynomial, and the rest supports the same factorial and time sum. The gamma-ratio estimate in Lemma 3.1 shows that fixed degree changes only the size polynomial. The reserved fractions and the admissible size base are fixed once; L is not chosen after R or q . At connected order $h = 1, 2$ the bound is $C_q \mu^{1-h} e^{-c_q R^2}$, uniformly in record times. Later independent births need not have compact spatial support.

For a field test u the exact second-cumulant identity is

$$\text{Var } \zeta_t^{\varepsilon, p}(u) = \int F_{1,t}^{\varepsilon, p} u^2 + \mu \int c_{2,t}^{\varepsilon, p}(u \otimes u).$$

The retained bounds at $u = q^{>R}$ and the restored errors prove (7.1). The same proof without a tail gives bounded second moments. Cauchy–Schwarz then proves two-time covariance cutoff approximation, uniformly in its two record times. $\square$

For $q \in \mathcal{Q}$, the true law inherits a marginal probability cutoff gate. Indeed $|q^{>R}| \leq |q|$, so (4.4) and the Holder estimate (4.8) compare the exact centerings uniformly in R, t . Off H6’s full-time exceptional event the raw paths agree. Chebyshev applied to (7.1), then this comparison, gives for every $\delta > 0$

$$\lim_{R \rightarrow \infty} \limsup_{\varepsilon \rightarrow 0} \sup_{t \leq T} \mathbb{P}\{|\zeta_t^\varepsilon(q^{>R})| > \delta\} = 0.$$

It is a marginal statement, not an assertion of a true covariance bound.

Uniform marginal variance alone does not imply a uniform path cutoff. For example, a randomly placed interval of length $1/m$ on the unit time circle has an indicator with variance at most $1/m$ at each fixed time, whereas its centered time supremum is $1 - 1/m$. Theorem 1.2 avoids that issue by (5.1) and the monotone speed-sum LLN. Likewise taking R beyond the actual polylog endpoint range at each fixed ε would produce a vacuous tail bound; the iterated order stated in the lemma is the relevant one.

For $q \in \mathcal{Q}$ the lemma also defines the Gaussian L^2 endpoint limit of $G_t(q\chi_R)$. Apply its bound to differences of compact cutoffs, pass to their fixed finite-dimensional Gaussian limits, and obtain an L^2 -Cauchy family. Joint probability comparison with the moment limit already proved identifies it as $Z_0(r_q(t))$. This supplies the indicated finite-dimensional covariance interpretation, not an all-polynomial strong-dual functional theorem.

7.1 The unresolved covariance transfer

On the bad event the mechanical bound allows a quadratic fluctuation difference of order $\varepsilon^{-1}N/\mu$. H1 and H6 give only

$$\varepsilon^{-2}\mathbb{E}[(N/\mu)^2 \mathbf{1}_{E_\varepsilon}] \leq C_p \varepsilon^{-2+(21/20)(1-2/p)}. \quad (7.2)$$

Its exponent approaches $-19/20$ as $p \rightarrow \infty$ and equals -1 at $p = 42$. Thus this method does not establish true prelimit covariance convergence. A growing upper bound supplies no lower bound on the actual covariance and proves no physical divergence.

A scalar example displays the logical gap. Take a bounded centered W and an independent event A_ε of probability $\varepsilon^{21/20}$, and set

$$X_\varepsilon = W + \varepsilon^{-1}(\mathbf{1}_{A_\varepsilon} - \varepsilon^{21/20}).$$

Then $X_\varepsilon - W \rightarrow 0$ in probability, its uncentered mean shift $\varepsilon^{1/20}$ vanishes, but

$$\text{Var } X_\varepsilon - \text{Var } W = \varepsilon^{-19/20}(1 - \varepsilon^{21/20}) \rightarrow \infty.$$

This is not a hard-sphere counterexample. Additional weighted true-flow control could prove covariance convergence; the present results leave it unresolved.

7.2 Other boundaries

The theorem identifies global quadratic moments and their first means. It does not establish a phase-space distribution limit of $\varepsilon^{-1}(\mathbb{E}\pi_t^\varepsilon - f_t)$, a full unit-activity mean corrector, or a local stress-corrector equation. No compact cubic activity result is used or reproduced. The invariant projection and Gaussian mapping are standard; the additional work is the clipped two-root weighted

record bound, its uniform weak-law use, the exact conserved closure, and the intrinsic corollary, all relative to the pinned package.

The compact initial position anchor is substantive for Lemma 7.1. With large initial position and small velocity, its implication $E \geq cR^2$ fails. A summable spatial Bol norm alone does not force a second spatial moment. The compact support assumption also makes the initial polynomial-to-compact substitution exact. Removal of this assumption requires separate weighted initial information.

A Exact source map and reproducibility

All imports are public text at immutable commits. F and V are in `mxym/math` at

87868bcb65bf0460a90d6bc2efc3324e46481230.

U is in `OpenAI/math` at

adc7f1241b42e322a6451854ab7e4b4c146bf78a.

The original shorthand `mxym/math009` did not identify an accessible repository; the audited source is entry 009 within `mxym/math`. The twenty-five audited F/V files have identical Git blob hashes at the first commit and `1c67b4146f9e54291c8a5ca1f42a028bf307629f`. No result here depends on an unpinned working tree.

A.1 Imported results and their exact locations

In the following map, paths are repository-relative and labels are the literal TeX labels in the pinned source. Line numbers are conveniences for the pinned bytes, not the basis of identification.

Virial supplement V. All of the following labels are in `notes/full-density-virial-stress/manuscript.tex`:

Use here	Exact source label(s)	Pinned line(s)
Model, diameter, ensemble	<code>ass:model</code> , <code>eq:ensemble</code>	46, 56
Collision and sphere normalization	<code>eq:collision</code> , <code>eq:J</code>	70, 82
H1	<code>imp:initial</code>	142
H2	<code>imp:algebra</code> , <code>eq:partition</code> , <code>eq:rootoutput</code>	145, 149, 151
H3	<code>imp:unweighted</code> , <code>eq:unweighted</code>	163, 165
H4	<code>imp:majorant</code> , <code>eq:majorant</code>	172, 179
H5	<code>imp:tree</code>	184
H6	<code>imp:coupling</code>	193
Mechanical identities	<code>lem:mechanics</code> , <code>eq:micro-virial</code> , <code>eq:micro-spread</code>	210, 213, 215
Vacancy expansion	<code>lem:initial</code> , <code>eq:vacancy</code>	233, 235
Incidence mean and clipping	<code>prop:incidence</code> , <code>eq:mark-bound</code> , <code>eq:clip</code> , <code>eq:rawbound</code> , <code>eq:scalar-moment</code>	257, 286, 296, 308, 323
First-moment transfer	<code>lem:transfer</code> , <code>eq:transfer</code> , <code>eq:true-incidence</code>	344, 346, 369

The proof of Lemma 3.2 reproduces the weighted mean argument in V. V's H_ε^p denotes an expectation; this manuscript's $\mathcal{H}_\varepsilon^p$ is random and $m_\varepsilon = \mathbb{E}\mathcal{H}_\varepsilon^p$. Confusing these two would bypass the required variance estimate.

Functional manuscript F. Use the prefix `preprints/009-functional-hard-sphere-fluctuations/v1/sections/`.

File	Exact labels	Use here
01_model.tex	ass:kinetic, lem:convention	Kinetic interval, path versions
02_imports.tex	eq:counts, prop:imports	Factorial/cumulant normalization and package
05_increments.tex	lem:remainder	Unweighted record errors
06_microscopic.tex	lem:record	Passive bounded records and zero continuation
07_functional.tex	thm:functional	Exactly centered Schwartz functional limit
08_tempered.tex	thm:tempered, eq:tempered-covariance	Strong-dual path limit, Gaussian covariance

The connected scaling at several roots in (3.6) uses F's normalized factorial convention and V's chronological birth/link count. The initial Gaussian covariance and initial/noise independence are those of F. Its positive-time test space is retained exactly as stated.

Upstream source U. Use the prefix `preprints/Hard-sphere-fluctuations-on-the-regular-Boltzmann-lifespan-September-23-2026/build/sections/`.

File	Exact labels or portion	Use here
02-initial-target.tex	prop:initial, proof's static cumulant bound	Bounded initial CLT and $\mu\ c_{2,0}\ _{TV} = O(\varepsilon)$
03-dhm-inputs.tex	Gaussian input envelope and dynamics	Bol inputs within H1
04-cluster-expansion.tex	Autonomous partition and coarse variation argument	Operational basis of H2/H3
05-cutoff.tex	prop:cutoff, eq:cutoff-probability	Full-time $O(\varepsilon^{21/20})$ probability comparison
06-sharp-asymptotics.tex	eq:sharp-input-bounds, eq:sharp-scaling, lem:sharp-factorials, lem:sharp-summability, eq:sharp-energy-sum	Chronological scaling, energy bins, Stirling and entropy sum
07-gaussian-limit.tex	Leading-tree cancellation and Gaussian identification	Basis of H5 and input F

The seven-file U inventory includes its `build/main.tex`. These operational and functional inputs are the conditional foundation; this manuscript does not claim to reprove the entire upstream hierarchy.

A.2 Proof contribution and conclusion ledger

Lemma 3.1 proves the previously missing degree-two, two-root passive-speed estimate by global clipping and retained Gaussian energy. The exact variance formula then proves (3.3); monotonicity proves the uniform pasted L^2 LLN and H6 gives the true probability LLN. Sections 4 and 5 supply maximal quadratic closure, the full thirteen-component theorem, the joint-field argument, and pasted covariance convergence. Corollary 6.1 adds the per-particle intrinsic observables via explicit differentials. Lemma 7.1 is a separate endpoint covariance result. The true prelimit covariance obstruction in (7.2) remains an unresolved transfer problem.

A.3 Reproduction package

The source archive contains this manuscript and all its sections, the PDF, `build.py`, `README.txt`, optimization-safe symbolic checks, negative-control evidence, and source provenance. The thirty-two exact pinned verification-cache files are bundled under `sources/`; their URL, byte count, SHA-256 and Git blob SHA-1 are recorded in `provenance/source-manifest.json` and `provenance/upstream-manifest.json`. This cache contains the audited imports, rather than complete standalone distributions of F, V and U. The public strict checker verifies all cache bytes and literal source labels offline; optional fetching uses ordinary TLS verification.

`audit/AUDIT.txt` is a documented public derivative of the original mathematical audit. The seven original audit objects remain unchanged and are represented by their hashes in `provenance/original-object-pins.json`. The original patch has SHA-256

6c659d81faafc9a803efe97da399e16aa7aa642ab352213e5c3f77d360442528.

The public checker uses explicit exceptions and verifies fixed audit, provenance, manuscript and cached-source anchors under both ordinary Python and `python -O`. Deliberate corruption is rejected in both modes. These checks verify identities, normalizations and source correspondence; the analytic proofs are in the text above. No Lean verification is claimed.

The build fixes the source epoch, omits volatile PDF dates and identifiers, disables shell escape, and performs three pdfLaTeX passes in a fresh directory. `build/determinism.json` records two fresh matching builds. The isolated fallback format loads `english hyphen.tex` when the standard installation lacks its generated format or font map. Byte reproducibility across different TeX or font versions is not asserted. `PUBLIC_FILES.json` is the publication whitelist; `package-manifest.json` hashes its files except the self-excluded manifest. `checks/verify_package.py` also rejects unexpected files.

References

- [1] *Functional hard-sphere fluctuations*, entry 009, version 1. [Public source at the pinned mxym/math commit](#).
- [2] *Full-density virial and spatial-spread corrections and the collisional momentum-balance defect*, version 1, 7 October 2026. [Public source at the same pinned commit](#).
- [3] *Hard-sphere fluctuations on the regular Boltzmann lifespan*, OpenAI Math Release, 23 September 2026. [Public TeX snapshot at the pinned OpenAI/math commit](#).